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### **BACKSIDE ISOLATION PILLAR REMOVAL FOR CAPACITANCE REDUCTION**

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#### **Abstract**

A semiconductor device is provided and includes first and second active regions, a shared gate structure disposed across the first and second active regions and including first and second metal gate material in the first and second active regions, respectively, a self-aligned backside dielectric pillar interposed between a first portion of the first metal gate material and a first portion of the second metal gate material and a shared gate plug interposed between a second portion of the first metal gate material and a second portion of the second metal gate material. The self-aligned backside dielectric pillar defines a pinched-off air gap extending across a width-wise portion of the self-aligned backside dielectric pillar and along a height-wise portion of the self-aligned backside dielectric pillar.

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## Background/Summary

### BACKGROUND

[0001] The present disclosure generally relates to fabrication methods and resulting structures for semiconductor devices. More specifically, the present disclosure relates backside isolation pillar removal for capacitance reduction for shared and non-shared gates for semiconductor devices.

[0002] A transistor is a semiconductor device used to amplify or switch electrical signals and power and is one of the basic building blocks of modern electronics. A field-effect transistor (FET) is a type of transistor that uses an electric field to control the flow of current in a semiconductor. An FET has three terminals: a source, a gate and a drain. FETs control the flow of current by the application of a voltage to the gate, which in turn alters the conductivity between the drain and the source.

### SUMMARY

[0003] According to an aspect of the disclosure, a semiconductor device is provided and includes first and second active regions, a shared gate structure disposed across the first and second active regions and including first and second metal gate material in the first and second active regions, respectively, a self-aligned backside dielectric pillar interposed between a first portion of the first metal gate material and a first portion of the second metal gate material and a shared gate plug interposed between a second portion of the first metal gate material and a second portion of the second metal gate material. The self-aligned backside dielectric pillar defines a pinched-off air gap extending across a width-wise portion of the self-aligned backside dielectric pillar and along a height-wise portion of the self-aligned backside dielectric pillar. In additional or alternative embodiments, the pinched-off air gap reduces parasitic capacitance between the first and second metal gate materials.

[0004] According to an aspect of the disclosure, a semiconductor device is provided and includes first and second active regions, a non-shared gate structure disposed across the first and second active regions and including first and second metal gate material in the first and second active regions, respectively, and a self-aligned backside dielectric pillar interposed between the first metal gate material and the second metal gate material. The backside dielectric pillar defines a pinched-off air gap extending across a width-wise portion of the self-aligned backside dielectric pillar and along a height-wise portion of the self-aligned backside dielectric pillar. In additional or alternative embodiments, the pinched-off air gap reduces parasitic capacitance between the first and second metal gate materials.

[0005] According to an aspect of the disclosure, a semiconductor device fabrication method is provided and includes forming a gate structure across first and second active regions and including first and second metal gate material in the first and second active regions, respectively, and first and second support structures underlying the first and second metal gate material, respectively, removing material from between the first and second metal gate material and from between the first and second support structures to form first and second openings, respectively, executing atomic layer deposition (ALD) to partially fill the first opening with dielectric defining a pinched-off air gap therein and to partially fill the second opening with the dielectric defining a partially pinched-off air gap therein and executing additional ALD to completely fill the partially pinched-off air gap with the dielectric whereby the dielectric forms a self-aligned backside dielectric pillar. In additional or alternative embodiments, the pinched-off air gap reduces parasitic capacitance between the first and second metal gate materials.

[0006] Additional technical features and benefits are realized through the techniques of the present disclosure. Embodiments and aspects of the disclosure are described in detail herein and are

considered a part of the claimed subject matter. For a better understanding, refer to the detailed description and to the drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The specifics of the exclusive rights described herein are particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other features and advantages of the embodiments of the disclosure are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

[0008] FIG. 1 is a top-down view of a semiconductor device in accordance with one or more embodiments;

[0009] FIG. 2A is a side view of the semiconductor device of FIG. 1 in accordance with one or more embodiments;

[0010] FIG. 2B is a side view of an enlarged portion of the semiconductor device of FIG. 2A in accordance with one or more embodiments;

[0011] FIG. 3A is a side view of the semiconductor device of FIG. 1 in accordance with one or more embodiments;

[0012] FIG. 3B is a side view of an enlarged portion of the semiconductor device of FIG. 3A in accordance with one or more embodiments;

[0013] FIG. 4 is a flow diagram illustrating a semiconductor device fabrication method in accordance with one or more embodiments;

[0014] FIG. 5 is a side view of an initial semiconductor device assembly in accordance with one or more embodiments;

[0015] FIG. 6 is a side view of a second semiconductor device assembly in accordance with one or more embodiments;

[0016] FIG. 7 is a side view of a third semiconductor device assembly in accordance with one or more embodiments; and

[0017] FIG. 8 is a side view of a fourth semiconductor device assembly in accordance with one or more embodiments.

[0018] The diagrams depicted herein are illustrative. There can be many variations to the diagram or the operations described therein without departing from the spirit of the disclosure. For instance, the actions can be performed in a differing order or actions can be added, deleted or modified. Also, the term “coupled” and variations thereof describes having a communications path between two elements and does not imply a direct connection between the elements with no intervening elements/connections between them. All of these variations are considered a part of the specification.

[0019] In the accompanying figures and following detailed description of the described embodiments, the various elements illustrated in the figures are provided with two or three digit reference numbers. With minor exceptions, the leftmost digit(s) of each reference number correspond to the figure in which its element is first illustrated.

### DETAILED DESCRIPTION

[0020] According to an aspect of the disclosure, a semiconductor device is provided and includes first and second active regions, a shared gate structure disposed across the first and second active regions and including first and second metal gate material in the first and second active regions, respectively, a self-aligned backside dielectric pillar interposed between a first portion of the first metal gate material and a first portion of the second metal gate material and a shared gate plug interposed between a second portion of the first metal gate material and a second portion of the second metal gate material. The self-aligned backside dielectric pillar defines a pinched-off air gap

extending across a width-wise portion of the self-aligned backside dielectric pillar and along a height-wise portion of the self-aligned backside dielectric pillar. In additional or alternative embodiments, the pinched-off air gap reduces parasitic capacitance between the first and second metal gate materials.

[0021] In accordance with additional or alternative embodiments, the shared gate structure includes first and second gate structures in the first and second active regions, the first gate structure includes first nanosheets surrounded by the first metal gate material and the second gate structure includes second nanosheets surrounded by the second metal gate material. The semiconductor device can therefore be provided as a field effect transistor (FET) device.

[0022] In accordance with additional or alternative embodiments, the semiconductor device further includes interlayer dielectric (ILD) extending across the first and second gate structures, a back-end-of-line (BEOL) layer disposed on the ILD, a frontside contact extending through the ILD and being electrically connected to the shared gate plug and the BEOL layer and a carrier layer disposed on the BEOL layer. The semiconductor device is therefore complementary-metal-oxide-semiconductor (CMOS) compatible.

[0023] In accordance with additional or alternative embodiments, the semiconductor device further includes a backside power rail (BPR) and a backside contact electrically connected to the BPR. The semiconductor device therefore provides for connection with a backside power distribution network (BSPDN).

[0024] In accordance with additional or alternative embodiments, the pinched-off air gap is ovular and elongate height-wise and contributes to reduction of parasitic capacitance between the first and second metal gate materials more than in a vertical direction.

[0025] In accordance with additional or alternative embodiments, the self-aligned backside dielectric pillar includes an upper pinched-off section delimiting an upper extent of the pinched-off air gap and proximate to the shared gate plug and a lower pinched-off section delimiting a lower extent of the pinched-off air gap and proximate to a lowermost plane of the first and second metal gate materials in the first and second active regions, respectively. The upper and lower pinched-off sections can therefore be formed as a direct result of atomic layer deposition (ALD) processing.

[0026] In accordance with additional or alternative embodiments, the side sections of the self-aligned backside dielectric pillar have a minimum width corresponding to a maximum width of the pinched-off air gap and respective widths of the side sections are up to about 20-33% of a total width of the backside dielectric pillar. The pinched-off air gap therefore occupies a significant fraction of the volume of the backside dielectric pillar.

[0027] According to an aspect of the disclosure, a semiconductor device is provided and includes first and second active regions, a non-shared gate structure disposed across the first and second active regions and including first and second metal gate material in the first and second active regions, respectively, and a self-aligned backside dielectric pillar interposed between the first metal gate material and the second metal gate material. The backside dielectric pillar defines a pinched-off air gap extending across a width-wise portion of the self-aligned backside dielectric pillar and along a height-wise portion of the self-aligned backside dielectric pillar. In additional or alternative embodiments, the pinched-off air gap reduces parasitic capacitance between the first and second metal gate materials.

[0028] In accordance with additional or alternative embodiments, the non-shared gate structure includes first and second gate structures in the first and second active regions, the first gate structure including first nanosheets surrounded by the first metal gate material and the second gate structure including second nanosheets surrounded by the second metal gate material. The semiconductor device can therefore be provided as a field effect transistor (FET) device.

[0029] In accordance with additional or alternative embodiments, the semiconductor device further includes interlayer dielectric (ILD) extending across the first and second gate structures, a back-end-of-line (BEOL) layer disposed on the ILD and a carrier layer disposed on the BEOL layer. The

semiconductor device is therefore complementary-metal-oxide-semiconductor (CMOS) compatible.

[0030] In accordance with additional or alternative embodiments, the semiconductor device further includes a backside power rail (BPR) and a backside contact electrically connected to the BPR. The semiconductor device therefore provides for connection with a backside power distribution network (BSPDN).

[0031] In accordance with additional or alternative embodiments, the pinched-off air gap is ovalar and elongate height-wise and contributes to reduction of parasitic capacitance between the first and second metal gate materials more than in a vertical direction.

[0032] In accordance with additional or alternative embodiments, the self-aligned backside dielectric pillar includes an upper pinched-off section delimiting an upper extent of the pinched-off air gap and proximate an uppermost plane of the first and second metal gate materials in the first and second active regions, respectively, and a lower pinched-off section delimiting a lower extent of the pinched-off air gap and proximate to a lowermost plane of the first and second metal gate materials in the first and second active regions, respectively. The upper and lower pinched-off sections can therefore be formed as a direct result of atomic layer deposition (ALD) processing.

[0033] In accordance with additional or alternative embodiments, side sections of the self-aligned backside dielectric pillar have a minimum width corresponding to a maximum width of the pinched-off air gap and respective widths of the side sections are up to about 20-33% of a total width of the backside dielectric pillar. The pinched-off air gap therefore occupies a significant fraction of the volume of the backside dielectric pillar.

[0034] According to an aspect of the disclosure, a semiconductor device fabrication method is provided and includes forming a gate structure across first and second active regions and including first and second metal gate material in the first and second active regions, respectively, and first and second support structures underlying the first and second metal gate material, respectively, removing material from between the first and second metal gate material and from between the first and second support structures to form first and second openings, respectively, executing atomic layer deposition (ALD) to partially fill the first opening with dielectric defining a pinched-off air gap therein and to partially fill the second opening with the dielectric defining a partially pinched-off air gap therein and executing additional ALD to completely fill the partially pinched-off air gap with the dielectric whereby the dielectric forms a self-aligned backside dielectric pillar. In additional or alternative embodiments, the pinched-off air gap reduces parasitic capacitance between the first and second metal gate materials.

[0035] In accordance with additional or alternative embodiments, the gate structure is one of a shared gate structure and a non-shared gate structure with the pinched-off air gap being useful in both cases though somewhat taller in the non-shared case.

[0036] In accordance with additional or alternative embodiments, the pinched-off air gap is ovalar and elongate height-wise and contributes to reduction of parasitic capacitance between the first and second metal gate materials more than in a vertical direction.

[0037] In accordance with additional or alternative embodiments, the executing of the ALD includes executing the ALD to form an upper pinched-off section delimiting an upper extent of the pinched-off air gap and proximate an uppermost plane of the first and second gate metal materials in the first and second active regions, respectively, and executing the ALD to form a lower pinched-off section delimiting a lower extent of the pinched-off air gap and proximate to a lowermost plane of the first and second gate metal materials in the first and second active regions, respectively. The upper and lower pinched-off sections can be formed as a direct result of atomic layer deposition (ALD) processing.

[0038] In accordance with additional or alternative embodiments, the executing of the ALD includes executing the ALD such that side sections of the self-aligned backside dielectric pillar have a minimum width corresponding to a maximum width of the pinched-off air gap and the

executing of the ALD includes executing the ALD such that respective widths of the side sections are up to about 20-33% of a total width of the backside dielectric pillar. The pinched-off air gap therefore occupies a significant fraction of the volume of the backside dielectric pillar.

[0039] In accordance with additional or alternative embodiments, the semiconductor device fabrication method further includes completing complementary-metal-oxide-semiconductor (CMOS) operations and completing backside power rail (BPR) formations. The semiconductor device is therefore complementary-metal-oxide-semiconductor (CMOS) compatible and provides for connection with a backside power distribution network (BSPDN).

[0040] For the sake of brevity, conventional techniques related to semiconductor device and integrated circuit (IC) fabrication may or may not be described in detail herein. Moreover, the various tasks and process steps described herein can be incorporated into a more comprehensive procedure or process having additional steps or functionality not described in detail herein. In particular, various steps in the manufacture of semiconductor devices and semiconductor-based ICs are well known and so, in the interest of brevity, many conventional steps will only be mentioned briefly herein or will be omitted entirely without providing the well-known process details.

[0041] Turning now to an overview of technologies that are more specifically relevant to aspects of the disclosure, certain semiconductor device structures are characterized as having a shared metal gate with a backside dielectric pillar for isolation between neighboring gate structures. In these or other similar cases, there can be a parasitic capacitance between metal gate materials of the neighboring gate structures across the backside dielectric pillar when the metal gate materials of the neighboring gate structures are relatively close to one another. This can be due to the incidence of defects at interfaces between the metal gate materials of the neighboring gate structures and the backside dielectric pillar.

[0042] An airgap formed in the backside dielectric pillar between the metal gate material of the neighboring gate structures can reduce parasitic capacitance issues due to the fact that dielectric materials, such as air in air gaps, tend not to dissipate charge. It is often difficult, however, to form airgaps in the backside dielectric pillar between metal gate material of the neighboring gate structures.

[0043] Turning now to an overview of the aspects of the disclosure, one or more embodiments of the disclosure address the above-described shortcomings of the prior art by providing a semiconductor device with an airgap, which is formed within a backside dielectric pillar between metal gate materials of neighboring gate structures, and S/D self-aligned dielectric isolation in order to reduce parasitic capacitance. In addition, processes are provided to enable backside self-aligned shallow trench isolation (STI) cut and dielectric fill operations for the airgap formation within dielectric isolation provided by the backside dielectric pillar.

[0044] The above-described aspects of the disclosure address the shortcomings of the prior art by providing for a semiconductor device that includes first and second active regions, a shared gate structure disposed across the first and second active regions and including first and second metal gate material in the first and second active regions, respectively, a self-aligned backside dielectric pillar interposed between a first portion of the first metal gate material and a first portion of the second metal gate material and a shared gate plug interposed between a second portion of the first metal gate material and a second portion of the second metal gate material. The self-aligned backside dielectric pillar defines a pinched-off air gap extending across a width-wise portion of the self-aligned backside dielectric pillar and along a height-wise portion of the self-aligned backside dielectric pillar.

[0045] In addition, the above-described aspects of the disclosure address the shortcomings of the prior art by providing for a semiconductor device that includes first and second active regions, a non-shared gate structure disposed across the first and second active regions and including first and second metal gate material in the first and second active regions, respectively, and a self-aligned backside dielectric pillar interposed between the first metal gate material and the second metal gate

material. The backside dielectric pillar defines a pinched-off air gap extending across a width-wise portion of the self-aligned backside dielectric pillar and along a height-wise portion of the self-aligned backside dielectric pillar.

[0046] Turning now to a more detailed description of aspects of the present disclosure, FIG. 1 depicts a top-down view of a semiconductor device **101**. FIGS. 2A and 2B and FIGS. 3A and 3B are side views of the semiconductor device **101** each in accordance with one or more embodiments and each taken from a perspective defined by cross-sectional view Y1 of FIG. 1. FIGS. 5-8 are side views of the semiconductor device **101** in various stages of assembly in accordance with one or more alternative embodiments and are each taken from the perspective defined by cross-sectional view Y1 of FIG. 1.

[0047] As shown in FIG. 1, the semiconductor device **101** includes first and second active regions **110** and **111** and first gate region **121**, second gate region **122** and third gate region **123**. The first gate region **121** is disposed across the first and second active regions **110** and **111**, the second gate region **122** is disposed across the first and second active regions **110** and **111** and the third gate region **123** is disposed across the first and second active regions **110** and **111**.

[0048] As shown in FIGS. 1, 2A and 2B (FIGS. 2A and 2B relate to the second gate region **122** of FIG. 1) and in accordance with one or more embodiments, the second gate region **122** of the semiconductor device **101** can be characterized as a shared gate structure **201** that is disposed across the first and second active regions **110** and **111**. The shared gate structure **201** includes a first gate structure **210** in the first active region **110** and a second gate structure **220** in the second active region **111**. The first gate structure **210** in the first active region **110** includes first metal gate material **211**, first nanosheets **212** surrounded by the first metal gate material **211** and a first gate support structure **213** underlying the first metal gate material **211**. The second gate structure **220** in the second active region **111** includes second metal gate material **221**, second nanosheets **222** surrounded by the second metal gate material **221** and a second gate support structure **223** underlying the second metal gate material **221**.

[0049] The semiconductor device **101** further includes a self-aligned backside dielectric pillar **230**, which is interposed between a first (i.e., lower) portion of the first metal gate material **211** and a first (i.e., lower) portion of the second metal gate material **221** and between the first and second gate support structures **213** and **223**, and a shared gate plug **240**. The shared gate plug **240** is disposed on the self-aligned backside dielectric pillar **230** and is interposed between a second (i.e., upper) portion of the first metal gate material **211** and a second (i.e., upper) portion of the second metal gate material **221**. In addition, the semiconductor device **101** also includes interlayer dielectric (ILD) **250** extending across the first and second gate structures **210** and **220**, a back-end-of-line (BEOL) layer **260** disposed on the ILD **250**, a frontside contact **270** extending through the ILD **250**, a carrier layer **280**, a backside power rail (BPR) **290** and a backside contact **295**. The frontside contact **270** is electrically connected at opposite ends thereof to the shared gate plug **240** and to the BEOL layer **260**. The carrier layer **280** is disposed on the BEOL layer **260**. The BPR **290** can include or be provided as a backside power delivery network (BSPDN). The backside contact **295** is electrically connected at opposite ends thereof to the BPR **290** and to source/drain (S/D) epitaxy associated with one of the first and second gate structures **210** and **220**.

[0050] The self-aligned backside dielectric pillar **230** is formed to define a pinched-off air gap **235**. The pinched-off air gap **235** extends in a width dimension across a width-wise portion of the self-aligned backside dielectric pillar **230** and extends in a height dimension along a height-wise portion of the self-aligned backside dielectric pillar **230**. In accordance with one or more embodiments, the pinched-off air gap **235** can be ovular and elongate height-wise (i.e., longer in the height dimension than the width dimension).

[0051] In order to form the pinched-off air gap **235**, the self-aligned backside dielectric pillar **230** includes an upper pinched-off section **236** and a lower pinched-off section **237**. The upper pinched-off section **236** delimits an upper extent of the pinched-off air gap **235** and is disposed proximate to

the shared gate plug **240**. The lower pinched-off section **237** delimits a lower extent of the pinched-off air gap **235** and is disposed proximate to a lowermost plane P of the first and second gate materials **211** and **221** in the first and second active regions **110** and **111**, respectively. The self-aligned backside dielectric pillar **230** also includes side sections **238**. The side sections **238** are curved and form sidewalls of the pinched-off air gap **235**. The side sections **238** can have a height-wise location of minimum width that corresponds to a height-wise location of maximum width of the pinched-off air gap **235**. Also, respective widths of the side sections **238** may be up to about 20-33% of a total width of the backside dielectric pillar **230** (i.e., the side sections **238** can be about 2 nm wide).

[0052] With the self-aligned backside dielectric pillar **230** being formed to define the pinched-off air gap **235**, the pinched-off air gap **235** provides for a reduced parasitic capacitance of the semiconductor device **101**. That is, even where there may be defects in the interfaces between the first and second metal gate materials **211** and **221** with the self-aligned backside dielectric pillar **230**, the pinched-off air gap **235** will tend not to dissipate stored charges. An overall performance of the semiconductor device **101** will be improved thereby.

[0053] It is to be understood that respective heights of the upper pinched-off section **236** and the lower pinched-off section **237** are controllable in some cases. For example, in an event the semiconductor device **101** exhibits some parasitic capacitance along the self-aligned backside dielectric pillar **230** in the height-wise dimension, the respective heights of the upper pinched-off section **236** and the lower pinched-off section **237** can be decreased so as to correspondingly increase the height of the pinched-off air gap **235**.

[0054] As shown in FIGS. **1**, **3A** and **3B** (FIGS. **3A** and **3B** relate to the second gate region **122** of FIG. **1**) and in accordance with one or more embodiments, the second gate region **122** of the semiconductor device **101** can be characterized as a non-shared gate structure **301** that is disposed across the first and second active regions **110** and **111**. The non-shared gate structure **301** includes a first gate structure **310** in the first active region **110** and a second gate structure **320** in the second active region **111**. The first gate structure **310** in the first active region **110** includes first metal gate material **311**, first nanosheets **312** surrounded by the first metal gate material **311** and a first gate support structure **313** (which will be described further below) underlying the first metal gate material **311**. The second gate structure **320** in the second active region **111** includes second metal gate material **321**, second nanosheets **322** surrounded by the second metal gate material **321** and a second gate support structure **323** (which will be described further below) underlying the second metal gate material **321**.

[0055] The semiconductor device **101** further includes a self-aligned backside dielectric pillar **330**, which is interposed between the first metal gate material **311** and the second metal gate material **321** and between the first and second gate support structures **313** and **323**. In addition, the semiconductor device **101** also includes interlayer dielectric (ILD) **350** extending across the first and second gate structures **310** and **320**, a back-end-of-line (BEOL) layer **360** disposed on the ILD **350**, a carrier layer **380**, a backside power rail (BPR) **390** and a backside contact **395**. The carrier layer **380** is disposed on the BEOL layer **360**. The BPR **390** can include or be provided as a backside power delivery network (BSPDN). The backside contact **395** is electrically connected at opposite ends thereof to the BPR **390** and to S/D epitaxy associated with one of the first and second gate structures **310** and **320**.

[0056] The self-aligned backside dielectric pillar **330** is formed to define a pinched-off air gap **335**. The pinched-off air gap **335** extends in a width dimension across a width-wise portion of the self-aligned backside dielectric pillar **330** and extends in a height dimension along a height-wise portion of the self-aligned backside dielectric pillar **330**. In accordance with one or more embodiments, the pinched-off air gap **335** can be ovular and elongate height-wise (i.e., longer in the height dimension than the width dimension).

[0057] In order to form the pinched-off air gap **335**, the self-aligned backside dielectric pillar **330**



includes an upper pinched-off section **336** and a lower pinched-off section **337**. The upper pinched-off section **336** delimits an upper extent of the pinched-off air gap **335** and is disposed proximate to an uppermost plane UP of the first and second gate materials **311** and **321** in the first and second active regions **110** and **111**, respectively. The lower pinched-off section **337** delimits a lower extent of the pinched-off air gap **335** and is disposed proximate to a lowermost plane LP of the first and second gate materials **311** and **321** in the first and second active regions **110** and **111**, respectively. The self-aligned backside dielectric pillar **330** also includes side sections **338**. The side sections **338** are curved and form sidewalls of the pinched-off air gap **335**. The side sections **338** can have a height-wise location of minimum width that corresponds to a height-wise location of maximum width of the pinched-off air gap **335**. Also, respective widths of the side sections **338** may be up to about 20-33% of a total width of the backside dielectric pillar **330** (i.e., the side sections **338** can be about 2 nm wide).

[0058] With the self-aligned backside dielectric pillar **330** being formed to define the pinched-off air gap **335**, the pinched-off air gap **335** provides for a reduced parasitic capacitance of the semiconductor device **101**. That is, even where there may be defects in the interfaces between the first and second metal gate materials **311** and **321** with the self-aligned backside dielectric pillar **330**, the pinched-off air gap **335** will tend not to dissipate stored charges. An overall performance of the semiconductor device **101** will be improved thereby.

[0059] It is to be understood that respective heights of the upper pinched-off section **336** and the lower pinched-off section **337** are controllable in some cases. For example, in an event the semiconductor device **101** exhibits some parasitic capacitance along the self-aligned backside dielectric pillar **330** in the height-wise dimension, the respective heights of the upper pinched-off section **336** and the lower pinched-off section **337** can be decreased so as to correspondingly increase the height of the pinched-off air gap **335**.

[0060] It is to be understood that, in general, a height of the pinched-off airgap **335** in the non-shared gate structure **301** will be greater than a height of the pinched-off airgap **235** in the shared gate structure **201**.

[0061] With reference to FIG. 4, a semiconductor device fabrication method **400** is provided to form the semiconductor device **101** of FIG. 1 and FIGS. 2A and 2B and/or the semiconductor device **101** of FIG. 1 and FIGS. 3A and 3B. As shown in FIG. 4, the semiconductor device fabrication method **400** includes forming a shared or non-shared gate structure across first and second active regions and including first and second metal gate material in the first and second active regions, respectively, and first and second gate support structures underlying the first and second metal gate material, respectively (block **401**), removing material from between the first and second metal gate material and from between the first and second support structures to form first and second openings, respectively (block **402**) executing atomic layer deposition (ALD) to partially fill the first opening with dielectric defining a pinched-off air gap therein, which is ovular and elongate height-wise, and to partially fill the second opening with the dielectric defining a partially pinched-off air gap therein (block **403**) and executing additional ALD to completely fill the partially pinched-off air gap with the dielectric whereby the dielectric forms a self-aligned backside dielectric pillar (block **404**). The semiconductor device fabrication method **400** can further include completing complementary-metal-oxide-semiconductor (CMOS) operations (block **4015**) and completing BPR formations (block **4045**).

[0062] The executing of the ALD of block **403** can include executing the ALD to form an upper pinched-off section delimiting an upper extent of the pinched-off air gap and proximate an uppermost plane of the first and second gate materials in the first and second active regions, respectively (block **4031**) and executing the ALD to form a lower pinched-off section delimiting a lower extent of the pinched-off air gap and proximate to a lowermost plane of the first and second gate materials in the first and second active regions, respectively (block **4032**). In addition, the executing of the ALD of block **403** can include executing the ALD such that side sections of the

self-aligned backside dielectric pillar have a minimum width corresponding to a maximum width of the pinched-off air gap and such that respective widths of the side sections are up to about 20-33% of a total width of the backside dielectric pillar.

[0063] With continued reference to FIG. 4 and with additional reference to FIGS. 5-8, certain operations of the semiconductor device fabrication method **400** will now be described.

[0064] Prior to formation of the initial semiconductor device assembly **501** of FIG. 5 being formed, a starting substrate is provided with a first layer of a first semiconductor, a second layer of a second semiconductor disposed on the first layer, an epitaxial layer disposed on the second layer, a third semiconductor layer (SiGe55) disposed on the epitaxial layer and interleaved layers of the first and second semiconductor disposed on the third semiconductor layer. At this point, first gate (RX) patterning and ILD formation are executed followed by second RX patterning to form first and second gate regions and removal of the third semiconductor layer. Next, a bottom dielectric (SASI) layer is formed in each of the first and second gate regions from which the third semiconductor layer was removed, ILD is filled into a trench defined between the first and second gate regions and a fin reveal operation is executed to define shallow trench isolation (STI) regions. This is followed by sidewall SiGe spacer formations (by deposition and etching) to define gate extensions and subsequent self-aligned isolation deposition and etch back, hard mask removal and dummy gate formation. At this point, CMOS S/D regions are grown in the active regions between the gate regions and this is followed by replacement metal gate (RMG) formation, shared gate metal fill (in the shared gate case), additional ILD deposition, middle-of-line (MOL)/BEOL formation and carrier wafer bonding. Next, the first and second layers of the first and second semiconductors are etched, followed by cap formation under the first and second gate regions and self-aligned STI etch to arrive at the initial semiconductor device assembly **501** of FIG. 5.

[0065] As shown in FIG. 5, the initial semiconductor device assembly **501** includes first gate structure **510** in a first active region, second gate structure **520** in a second active region, first gate support structure **530** underlying the first gate structure **510** and second gate support structure **540** underlying the second gate structure **520**. The first gate structure **510** includes first nanosheets **511** surrounded by first metal gate material **512** and the second gate structure **520** includes second nanosheets **521** surrounded by second metal gate material **522**. Isolation dielectric **550** and a shared gate plug **555** are interposed between the first and second metal gate materials **512** and **522**. ILD **560** is disposed on the first and second metal gate materials **512** and **522**, BEOL layer **565** is disposed on the ILD **560** and a carrier wafer **570** is disposed on the BEOL layer **565**. A frontside contact **575** extends through the ILD **560** to electrically connect the BEOL layer **565** with the shared gate plug **555**. The first gate support structure **530** includes a bottom dielectric layer **531**, a pillar of silicon epitaxial material **532** and a cap **533**. The bottom dielectric layer **531**, the pillar of silicon epitaxial material **532** and the cap **533** are surrounded by a dielectric liner **534** and ILD material **535**. The second gate support structure **540** includes a bottom dielectric layer **541**, a pillar of silicon epitaxial material **542** and a cap **543**. The bottom dielectric layer **541**, the pillar of silicon epitaxial material **542** and the cap **543** are surrounded by a dielectric liner **544** and ILD material **545**.

[0066] As shown in FIG. 6, a second semiconductor device assembly **601** is provided following removal of the caps **533** and **543** of FIG. 5 and removal of the isolation dielectric **550** of FIG. 5 executed with respect to the initial semiconductor device assembly **501** of FIG. 5. The second semiconductor device assembly **601** is thus characterized as having a first opening **610** between the first and second metal gate materials **512** and **522** and a second opening **620** between the first and second gate support structures **530** and **540**.

[0067] As shown in FIG. 7, a third semiconductor device assembly **701** is provided following an ALD operation executed with respect to the second semiconductor device assembly **601** of FIG. 6. The ALD operation results in an ALD dielectric fill that fills the first opening **610** and the second opening **620** of FIG. 6 with a first dielectric pillar **710**. The ALD operation is isotropic and results

in the first dielectric pillar **710** including upper pinched-off section **711**, proximate to the shared gate plug **555** of FIG. 5 (or the ILD **560** of FIG. 5 in the non-shared gate case), first curved sidewalls **712** along the first and second metal gate materials **512** and **522**, lower pinched-off section **713** proximate to a lowermost plane P of the first and second metal gate materials **512** and **513** and second curved sidewalls **714** along the first and second gate support structures **530** and **540**. The ALD operation thus results in the first dielectric pillar **710** being formed to define a pinched-off air gap **720** between the first and second metal gate materials **512** and **513** and a partially pinched-off air gap **730** between the first and second gate support structures **530** and **540**. [0068] In accordance with embodiments, the first curved sidewalls **712** can have a width of about 20-33% (or about 2 nm) of a total width of the first opening **610**.

[0069] As shown in FIG. 8, a fourth semiconductor device assembly **801** is provided following an additional ALD operation executed with respect to the third semiconductor device assembly **701** of FIG. 7. The additional ALD operation results in an ALD dielectric fill that fills the partially pinched-off air gap **730** of FIG. 7 with dielectric material and thus transforms the first dielectric pillar **710** of FIG. 7 into a backside dielectric pillar **810** formed to define the pinched-off air gap **720**.

[0070] Following formation of the fourth semiconductor device assembly **801** of FIG. 8, the semiconductor device **101** of FIGS. 1, 2A and 2B (or the semiconductor device **101** of FIGS. 1, 3A and 3B) can be arrived at by multiple backside semiconductor manufacturing processes.

[0071] Various embodiments of the present disclosure are described herein with reference to the related drawings. Alternative embodiments can be devised without departing from the scope of this disclosure. Although various connections and positional relationships (e.g., over, below, adjacent, etc.) are set forth between elements in the following description and in the drawings, persons skilled in the art will recognize that many of the positional relationships described herein are orientation-independent when the described functionality is maintained even though the orientation is changed. These connections and/or positional relationships, unless specified otherwise, can be direct or indirect, and the present disclosure is not intended to be limiting in this respect.

Accordingly, a coupling of entities can refer to either a direct or an indirect coupling, and a positional relationship between entities can be a direct or indirect positional relationship. As an example of an indirect positional relationship, references in the present description to forming layer "A" over layer "B" include situations in which one or more intermediate layers (e.g., layer "C") is between layer "A" and layer "B" as long as the relevant characteristics and functionalities of layer "A" and layer "B" are not substantially changed by the intermediate layer(s).

[0072] The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms "comprises," "comprising," "includes," "including," "has," "having," "contains" or "containing," or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus.

[0073] Additionally, the term "exemplary" is used herein to mean "serving as an example, instance or illustration." Any embodiment or design described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The terms "at least one" and "one or more" are understood to include any integer number greater than or equal to one, i.e. one, two, three, four, etc. The terms "a plurality" are understood to include any integer number greater than or equal to two, i.e. two, three, four, five, etc. The term "connection" can include an indirect "connection" and a direct "connection."

[0074] References in the specification to "one embodiment," "an embodiment," "an example embodiment," etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment may or may not include the particular feature,

structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0075] For purposes of the description hereinafter, the terms “upper,” “lower,” “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” and derivatives thereof shall relate to the described structures and methods, as oriented in the drawing figures. The terms “overlying,” “atop,” “on top,” “positioned on” or “positioned atop” mean that a first element, such as a first structure, is present on a second element, such as a second structure, wherein intervening elements such as an interface structure can be present between the first element and the second element. The term “direct contact” means that a first element, such as a first structure, and a second element, such as a second structure, are connected without any intermediary conducting, insulating or semiconductor layers at the interface of the two elements.

[0076] Spatially relative terms, e.g., “beneath,” “below,” “lower,” “above,” “upper,” and the like, can be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” can encompass both an orientation of above and below. The device can be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0077] The phrase “selective to,” such as, for example, “a first element selective to a second element,” means that the first element can be etched and the second element can act as an etch stop.

[0078] The terms “about,” “substantially,” “approximately,” and variations thereof, are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of  $\pm 8\%$  or  $5\%$ , or  $2\%$  of a given value.

[0079] The term “conformal” (e.g., a conformal layer) means that the thickness of the layer is substantially the same on all surfaces, or that the thickness variation is less than 15% of the nominal thickness of the layer.

[0080] The terms “epitaxial growth and/or deposition” and “epitaxially formed and/or grown” mean the growth of a semiconductor material (crystalline material) on a deposition surface of another semiconductor material (crystalline material), in which the semiconductor material being grown (crystalline overlayer) has substantially the same crystalline characteristics as the semiconductor material of the deposition surface (seed material). In an epitaxial deposition process, the chemical reactants provided by the source gases can be controlled and the system parameters can be set so that the depositing atoms arrive at the deposition surface of the semiconductor substrate with sufficient energy to move about on the surface such that the depositing atoms orient themselves to the crystal arrangement of the atoms of the deposition surface. An epitaxially grown semiconductor material can have substantially the same crystalline characteristics as the deposition surface on which the epitaxially grown material is formed. For example, an epitaxially grown semiconductor material deposited on a  $\{100\}$  orientated crystalline surface can take on a  $\{100\}$  orientation. In some embodiments of the disclosure, epitaxial growth and/or deposition processes can be selective to forming on semiconductor surface, and cannot deposit material on exposed surfaces, such as silicon dioxide or silicon nitride surfaces.

[0081] As previously noted herein, for the sake of brevity, conventional techniques related to semiconductor device and integrated circuit (IC) fabrication may or may not be described in detail herein. By way of background, however, a more general description of the semiconductor device

fabrication processes that can be utilized in implementing one or more embodiments of the present disclosure will now be provided. Although specific fabrication operations used in implementing one or more embodiments of the present disclosure can be individually known, the described combination of operations and/or resulting structures of the present disclosure are unique. Thus, the unique combination of the operations described in connection with the fabrication of a semiconductor device according to the present disclosure utilize a variety of individually known physical and chemical processes performed on a semiconductor (e.g., silicon) substrate, some of which are described in the immediately following paragraphs.

[0082] In general, the various processes used to form a micro-chip that will be packaged into an IC fall into four general categories, namely, film deposition, removal/etching, semiconductor doping and patterning/lithography. Deposition is any process that grows, coats, or otherwise transfers a material onto the wafer. Available technologies include physical vapor deposition (PVD), chemical vapor deposition (CVD), electrochemical deposition (ECD), molecular beam epitaxy (MBE) and more recently, atomic layer deposition (ALD) among others. Removal/etching is any process that removes material from the wafer. Examples include etch processes (either wet or dry), and chemical-mechanical planarization (CMP), and the like. Semiconductor doping is the modification of electrical properties by doping, for example, transistor sources and drains, generally by diffusion and/or by ion implantation. These doping processes are followed by furnace annealing or by rapid thermal annealing (RTA). Annealing serves to activate the implanted dopants. Films of both conductors (e.g., poly-silicon, aluminum, copper, etc.) and insulators (e.g., various forms of silicon dioxide, silicon nitride, etc.) are used to connect and isolate transistors and their components. Selective doping of various regions of the semiconductor substrate allows the conductivity of the substrate to be changed with the application of voltage. By creating structures of these various components, millions of transistors can be built and wired together to form the complex circuitry of a modern microelectronic device. Semiconductor lithography is the formation of three-dimensional relief images or patterns on the semiconductor substrate for subsequent transfer of the pattern to the substrate. In semiconductor lithography, the patterns are formed by a light sensitive polymer called a photo-resist. To build the complex structures that make up a transistor and the many wires that connect the millions of transistors of a circuit, lithography and etch pattern transfer steps are repeated multiple times. Each pattern being printed on the wafer is aligned to the previously formed patterns and slowly the conductors, insulators and selectively doped regions are built up to form the final device.

[0083] The flowchart and block diagrams in the Figures illustrate possible implementations of fabrication and/or operation methods according to various embodiments of the present disclosure. Various functions/operations of the method are represented in the flow diagram by blocks. In some alternative implementations, the functions noted in the blocks can occur out of the order noted in the Figures. For example, two blocks shown in succession can, in fact, be executed substantially concurrently, or the blocks can sometimes be executed in the reverse order, depending upon the functionality involved.

[0084] The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments described. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments described herein.

## Claims

- 1.** A semiconductor device, comprising: first and second active regions; a shared gate structure disposed across the first and second active regions and comprising first and second metal gate material in the first and second active regions, respectively; a self-aligned backside dielectric pillar interposed between a first portion of the first metal gate material and a first portion of the second metal gate material; and a shared gate plug interposed between a second portion of the first metal gate material and a second portion of the second metal gate material, the self-aligned backside dielectric pillar defining a pinched-off air gap extending across a width-wise portion of the self-aligned backside dielectric pillar and along a height-wise portion of the self-aligned backside dielectric pillar.
- 2.** The semiconductor device according to claim 1, wherein: the shared gate structure comprises first and second gate structures in the first and second active regions, the first gate structure comprising first nanosheets surrounded by the first metal gate material, and the second gate structure comprising second nanosheets surrounded by the second metal gate material.
- 3.** The semiconductor device according to claim 2, further comprising: interlayer dielectric (ILD) extending across the first and second gate structures; a back-end-of-line (BEOL) layer disposed on the ILD; a frontside contact extending through the ILD and being electrically connected to the shared gate plug and the BEOL layer; and a carrier layer disposed on the BEOL layer.
- 4.** The semiconductor device according to claim 2, further comprising a backside power rail (BPR) and a backside contact electrically connected to the BPR.
- 5.** The semiconductor device according to claim 1, wherein the pinched-off air gap is ovular and elongate height-wise.
- 6.** The semiconductor device according to claim 1, wherein the self-aligned backside dielectric pillar comprises: an upper pinched-off section delimiting an upper extent of the pinched-off air gap and proximate to the shared gate plug; and a lower pinched-off section delimiting a lower extent of the pinched-off air gap and proximate to a lowermost plane of the first and second metal gate materials in the first and second active regions, respectively.
- 7.** The semiconductor device according to claim 1, wherein: side sections of the self-aligned backside dielectric pillar have a minimum width corresponding to a maximum width of the pinched-off air gap, and respective widths of the side sections are up to about 20-33% of a total width of the backside dielectric pillar.
- 8.** A semiconductor device, comprising: first and second active regions; a non-shared gate structure disposed across the first and second active regions and comprising first and second metal gate material in the first and second active regions, respectively; and a self-aligned backside dielectric pillar interposed between the first metal gate material and the second metal gate material, the backside dielectric pillar defining a pinched-off air gap extending across a width-wise portion of the self-aligned backside dielectric pillar and along a height-wise portion of the self-aligned backside dielectric pillar.
- 9.** The semiconductor device according to claim 8, wherein: the non-shared gate structure comprises first and second gate structures in the first and second active regions, the first gate structure comprising first nanosheets surrounded by the first metal gate material, and the second gate structure comprising second nanosheets surrounded by the second metal gate material.
- 10.** The semiconductor device according to claim 9, further comprising: interlayer dielectric (ILD) extending across the first and second gate structures; a back-end-of-line (BEOL) layer disposed on the ILD; and a carrier layer disposed on the BEOL layer.
- 11.** The semiconductor device according to claim 9, further comprising a backside power rail (BPR) and a backside contact electrically connected to the BPR.
- 12.** The semiconductor device according to claim 8, wherein the pinched-off air gap is ovular and elongate height-wise.
- 13.** The semiconductor device according to claim 8, wherein the self-aligned backside dielectric

pillar comprises: an upper pinched-off section delimiting an upper extent of the pinched-off air gap and proximate an uppermost plane of the first and second metal gate materials in the first and second active regions, respectively; and a lower pinched-off section delimiting a lower extent of the pinched-off air gap and proximate to a lowermost plane of the first and second metal gate materials in the first and second active regions, respectively.

**14.** The semiconductor device according to claim 8, wherein: side sections of the self-aligned backside dielectric pillar have a minimum width corresponding to a maximum width of the pinched-off air gap, and respective widths of the side sections are up to about 20-33% of a total width of the backside dielectric pillar.

**15.** A semiconductor device fabrication method, comprising: forming a gate structure across first and second active regions and comprising first and second metal gate material in the first and second active regions, respectively, and first and second support structures underlying the first and second metal gate material, respectively; removing material from between the first and second metal gate material and from between the first and second support structures to form first and second openings, respectively; executing atomic layer deposition (ALD) to partially fill the first opening with dielectric defining a pinched-off air gap therein and to partially fill the second opening with the dielectric defining a partially pinched-off air gap therein; and executing additional ALD to completely fill the partially pinched-off air gap with the dielectric whereby the dielectric forms a self-aligned backside dielectric pillar.

**16.** The semiconductor device fabrication method according to claim 15, wherein the gate structure is one of a shared gate structure and a non-shared gate structure.

**17.** The semiconductor device fabrication method according to claim 15, wherein the pinched-off air gap is ovular and elongate height-wise.

**18.** The semiconductor device fabrication method according to claim 15, wherein the executing of the ALD comprises: executing the ALD to form an upper pinched-off section delimiting an upper extent of the pinched-off air gap and proximate an uppermost plane of the first and second gate metal materials in the first and second active regions, respectively; and executing the ALD to form a lower pinched-off section delimiting a lower extent of the pinched-off air gap and proximate to a lowermost plane of the first and second gate metal materials in the first and second active regions, respectively.

**19.** The semiconductor device fabrication method according to claim 15, wherein: the executing of the ALD comprises executing the ALD such that side sections of the self-aligned backside dielectric pillar have a minimum width corresponding to a maximum width of the pinched-off air gap, and the executing of the ALD comprises executing the ALD such that respective widths of the side sections are up to about 20-33% of a total width of the backside dielectric pillar.

**20.** The semiconductor device fabrication method according to claim 15, further comprising: completing complementary-metal-oxide-semiconductor (CMOS) operations; and completing backside power rail (BPR) formations.

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